

Port pins P2.0 through P2.7 of IC1 are connected to seven-segment displays as shown in Fig. 1. To make the PCB layout easier, DIS1, DIS3 and DIS5 are connected differently than DIS2, DIS4 and DIS6. Pins e, d, c, dp, b, a, f and g of 7-segment displays DIS1, DIS3 and DIS5 are connected to port pins P2.0 through P2.7, respectively, whereas in case of DIS2, DIS4 and DIS6, pins dp, c, d, e, g, f, a and b are connected to port pins P2.0 through P2.7, respectively. This is actually a trade-off as simplicity of the PCB layout has increased the complexity of the scanning software.

Port P2 provides segment data for 7-segment displays DIS1 through

DIS6. Port pins P0.1 through P0.7 of IC1 are pulled up with a 10-kilo-ohm resistor network (RNW1). P0.2 through P0.7 are used to select the seven-segment displays. These pins drive the base of pnp transistors T7 through T2 to select the corresponding display. For example, if pin P0.2 of microcontroller AT89C52 goes low, transistor T7 is driven into saturation, which provides supply to the common-anode pin 3 or pin 8 of 7-segment display DIS6.

Microcontroller IC1 provides segment data and display-enabled signal simultaneously in time-division multiplexing mode for displaying a particular number on the seven-

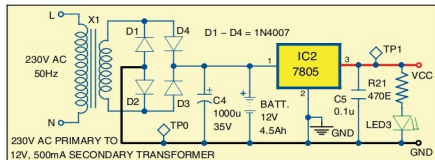


Fig. 2: Power supply section

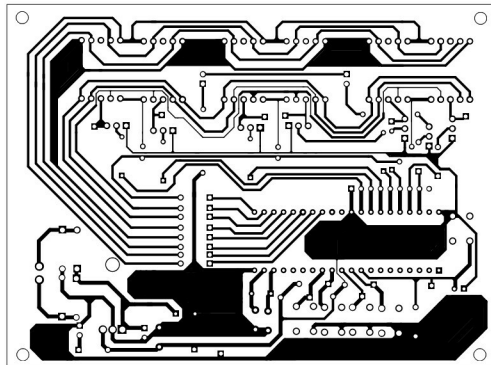


Fig. 3: An actual-size, single-side PCB for stop clock including the power supply section

## PARTS LIST

### Semiconductors:

|           |                                        |
|-----------|----------------------------------------|
| IC1       | - AT89C52 microcontroller              |
| IC2       | - 7805, 5V regulator                   |
| T1-T7     | - BC557 pnp transistor                 |
| D1-D4     | - 1N4007 rectifier diode               |
| DIS1-DIS6 | - LT542 common-anode 7-segment display |
| LED1-LED3 | - 5mm LED                              |

### Resistors (all ¼-watt, 5% carbon):

|         |                                |
|---------|--------------------------------|
| R1-R5   | - 10-kilo-ohm                  |
| R6-R13  | - 100-ohm                      |
| R14-R20 | - 1-kilo-ohm                   |
| R21     | - 470-ohm                      |
| RNW1    | - 10-kilo-ohm resistor network |

### Capacitors:

|        |                            |
|--------|----------------------------|
| C1     | - 10µF, 16V electrolytic   |
| C2, C3 | - 33pF ceramic disk        |
| C4     | - 1000µF, 35V electrolytic |
| C5     | - 0.1µF ceramic disk       |

### Miscellaneous:

|                                   |                                                       |
|-----------------------------------|-------------------------------------------------------|
| X1                                | - 230V AC primary to 12V, 500mA secondary transformer |
| X <sub>cr</sub> , X <sub>al</sub> | - 12MHz crystal                                       |
| S1-S5                             | - Push-to-on tactile switch                           |
| BATT.                             | - 12V, 4.5 Ah battery                                 |

segment display. Segment data and display-enabled pulse for the display are refreshed continuously at a very fast rate. Thus the display appears to be continuous due to persistence of vision even though the segment displays light up one by one. Port pin P0.1 is used to drive LED1 and LED2 so as to provide dots between DIS2

and DIS3, and DIS4 and DIS5, respectively.

Switches S1 through S4 are used for reset, start, stop and load. Port pins P3.0 through P3.3 are pulled up by resistors R1 through R4 and connected to switches S1 through S4, respectively. Switch S5 is used to manually reset the microcontroller, while the power-on reset signal for the microcontroller is derived from the combination of capacitor C1 and resistor R5. A 12MHz crystal is used to generate clock frequency for the microcontroller.

Fig. 2 shows the power supply circuit. The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of